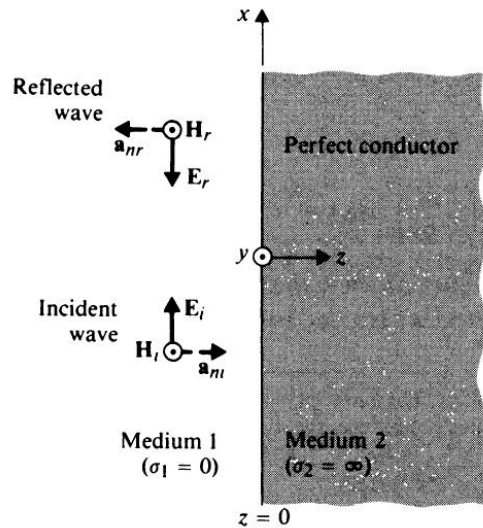


Revisit: Incidence at a Plane Conducting Boundary

Normal Incidence



$$\mathbf{E}_i(z) = \underline{\mathbf{a}_x} E_{i0} e^{-j\beta_1 z}$$

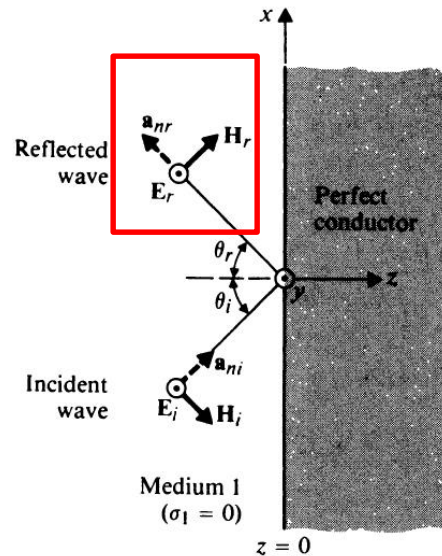
$$\mathbf{E}_{r0} = -\mathbf{E}_{i0}$$

$$\begin{aligned} \mathbf{E}_1(z) &= \mathbf{a}_x E_{i0} (e^{-j\beta_1 z} - e^{+j\beta_1 z}) \\ &= -\mathbf{a}_x j 2 E_{i0} \sin \beta_1 z. \end{aligned}$$

$$\mathbf{H}_1(z) = \mathbf{H}_i(z) + \mathbf{H}_r(z) = \mathbf{a}_y 2 \frac{E_{i0}}{\eta_1} \cos \beta_1 z.$$

$$\mathcal{P}_{av} = \frac{1}{2} \Re(\mathbf{E} \times \mathbf{H}^*) = 0$$

Perpendicular polarization



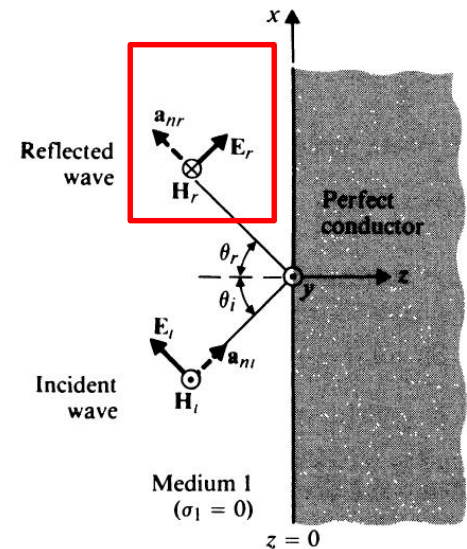
$$\mathbf{E}_i(x, z) = \underline{\mathbf{a}_y} E_{i0} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}$$

$$\mathbf{E}_r(x, z) = -\mathbf{a}_y E_{i0} e^{-j\beta_1(x \sin \theta_i - z \cos \theta_i)}.$$

$$\begin{aligned} \mathbf{E}_1(x, z) &= \mathbf{E}_i(x, z) + \mathbf{E}_r(x, z) \\ &= \mathbf{a}_y E_{i0} (e^{-j\beta_1 z \cos \theta_i} - e^{j\beta_1 z \cos \theta_i}) e^{-j\beta_1 x \sin \theta_i} \\ &= -\mathbf{a}_y j 2 E_{i0} \sin(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}. \end{aligned}$$

$$\begin{aligned} \mathbf{H}_1(x, z) &= -2 \frac{E_{i0}}{\eta_1} [\mathbf{a}_x \cos \theta_i \cos(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i} \\ &\quad + \mathbf{a}_z j \sin \theta_i \sin(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}]. \end{aligned}$$

Parallel polarization



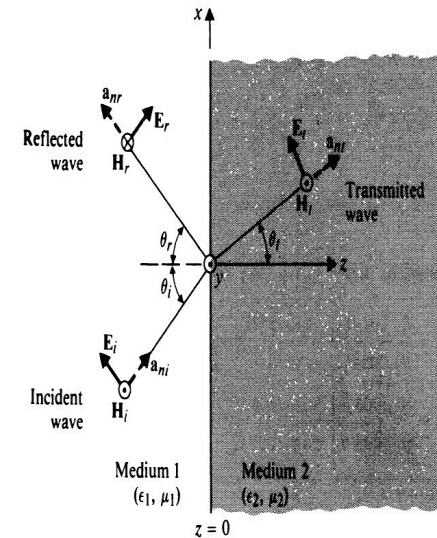
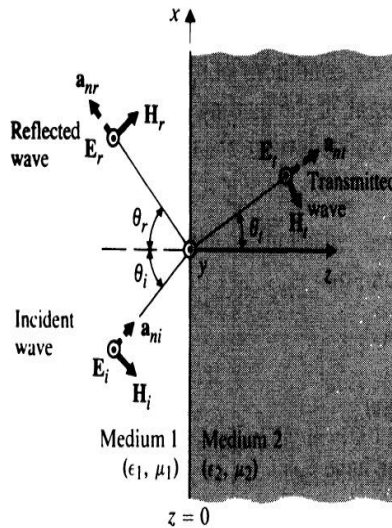
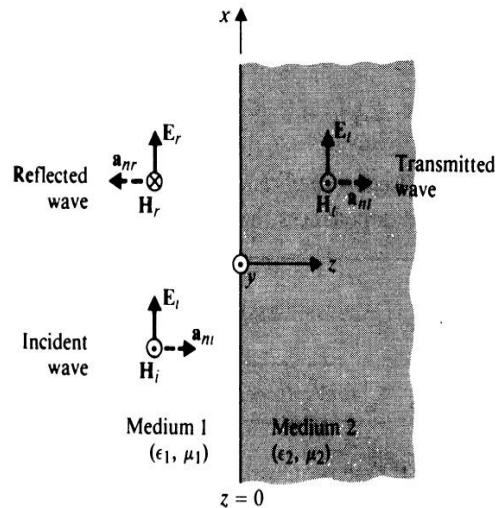
$$\mathbf{E}_i(x, z) = \underline{E_{i0}(\mathbf{a}_x \cos \theta_i - \mathbf{a}_z \sin \theta_i)} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}$$

$$\mathbf{E}_r(x, z) = -E_{i0}(\mathbf{a}_x \cos \theta_i + \mathbf{a}_z \sin \theta_i) e^{-j\beta_1(x \sin \theta_i - z \cos \theta_i)}$$

$$\begin{aligned} \mathbf{E}_1(x, z) &= \mathbf{E}_i(x, z) + \mathbf{E}_r(x, z) \\ &= -2 E_{i0} [\mathbf{a}_x j \cos \theta_i \sin(\beta_1 z \cos \theta_i) \\ &\quad + \mathbf{a}_z \sin \theta_i \cos(\beta_1 z \cos \theta_i)] e^{-j\beta_1 x \sin \theta_i}. \end{aligned}$$

$$\begin{aligned} \mathbf{H}_1(x, z) &= \mathbf{H}_i(x, z) + \mathbf{H}_r(x, z) \\ &= \mathbf{a}_y 2 \frac{E_{i0}}{\eta_1} \cos(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}. \end{aligned}$$

Incidence at a Plane Dielectric Boundary



$$\Gamma = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$\tau = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2}{\eta_2 + \eta_1}$$

$$1 + \Gamma = \tau$$

$$\begin{aligned} \Gamma_{\perp} &= \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} \\ &= \frac{(\eta_2 / \cos \theta_t) - (\eta_1 / \cos \theta_i)}{(\eta_2 / \cos \theta_t) + (\eta_1 / \cos \theta_i)} \end{aligned}$$

$$\begin{aligned} \tau_{\perp} &= \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} \\ &= \frac{2(\eta_2 / \cos \theta_t)}{(\eta_2 / \cos \theta_t) + (\eta_1 / \cos \theta_i)} \end{aligned}$$

$$1 + \Gamma_{\perp} = \tau_{\perp}$$

$$\Gamma_{\parallel} = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

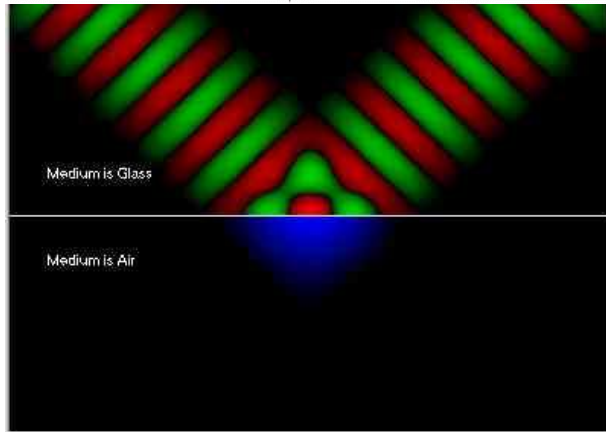
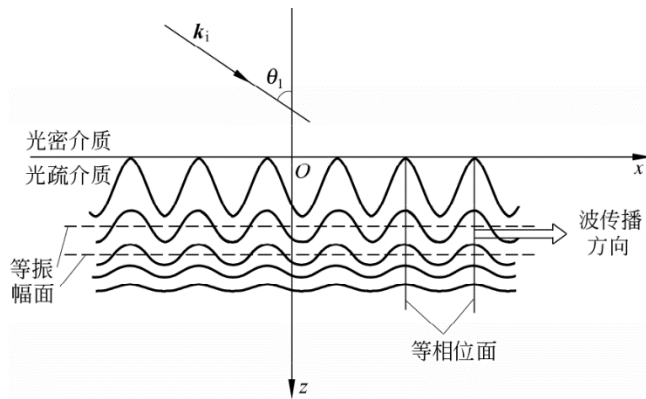
$$\tau_{\parallel} = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

$$1 + \Gamma_{\parallel} = \tau_{\parallel} \left(\frac{\cos \theta_t}{\cos \theta_i} \right)$$

Total reflection (全反射)

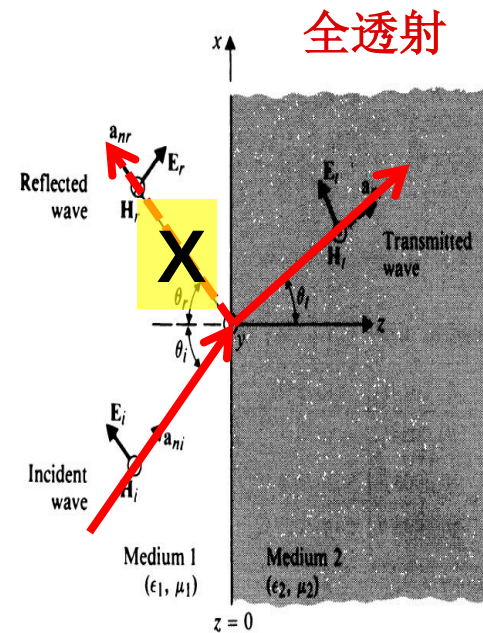
Critical angle ($\theta_2=90^\circ$):

$$\theta_c = \sin^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}} = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$



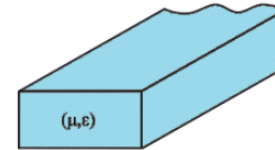
Brewster angle (布鲁斯特角) for parallel polarization

$$\theta_{B||} = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}} = \tan^{-1} \left(\frac{n_2}{n_1} \right)$$



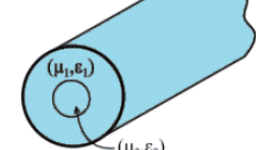


Rectangular 方形



Dielectric Slab Waveguide

Circular 圆形



Optical Fiber

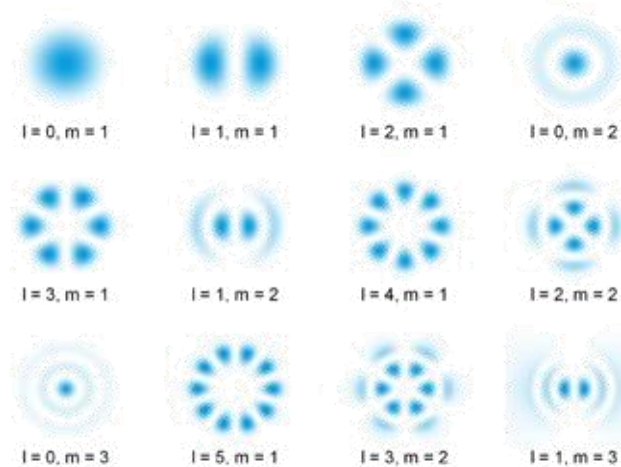
Waveguides

Metal
One **enclosed conductor** with an
insulating media

Dielectrics

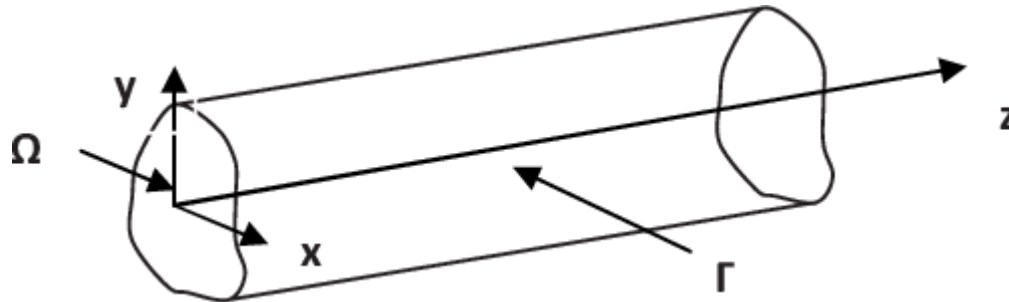
Must operate the waveguide at a frequency **above** the cutoff frequency
with dispersion 色散

mode: a trajectory of a physical system
which does not change shape as the system
evolves.

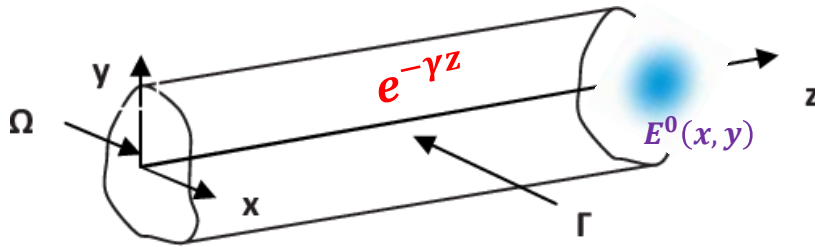


10-2

General wave behaviors along uniform Guiding structures



2D vector phasor and harmonic time dependence



all field components $E(x, y, z; t)$

Harmonic time dependence: $e^{j\omega t}$

propagation constant $\gamma = \alpha + j\beta$

the dependence on z for all field components as $e^{-\gamma z}$

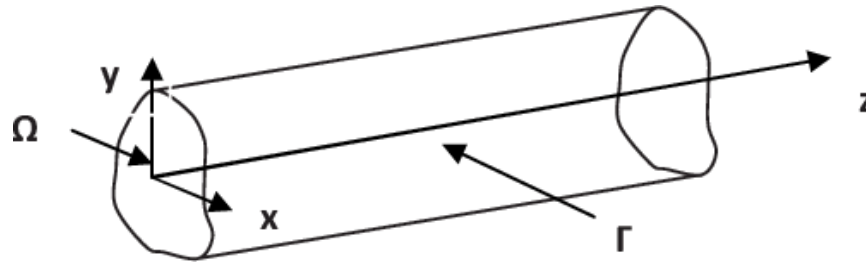
$$e^{-\gamma z} e^{j\omega t} = e^{-\alpha z} e^{j(\omega t - \beta z)}$$

$$\therefore E(x, y, z; t) = \text{Re}[E^0(x, y) e^{j(\omega t - \gamma z)}]$$

$$H(x, y, z; t) = \text{Re}[H^0(x, y) e^{j(\omega t - \gamma z)}]$$

Helmholtz's eqs

$$\nabla_T^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$



$$\nabla^2 E + k^2 E = 0$$

$$k^2 = \omega^2 \mu \epsilon$$

k – wavenumber

- We separate the Laplacian operator into 2 parts:

$$e^{-\gamma z} e^{j\omega t}$$

$$\gamma = \alpha + j\beta$$

$$\nabla^2 E = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) E = (\nabla_T^2 + \nabla_z^2) E = \left(\nabla_T^2 + \frac{\partial^2}{\partial z^2} \right) E = (\nabla_T^2 + \gamma^2) E$$

$$\therefore \nabla_T^2 E + (k^2 + \gamma^2) E = 0$$

$$\nabla_T^2 E + h^2 E = 0$$

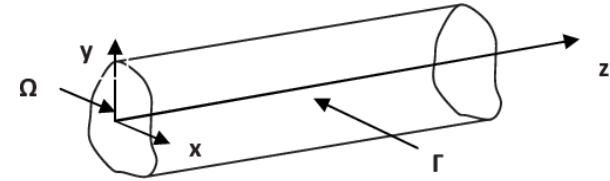
Plane wave

$$\nabla_T^2 E = 0$$

$$\begin{aligned} h^2 &= k^2 + \gamma^2 \\ &= k^2 - \beta^2 \end{aligned}$$

Getting the “ T part” from “ z part”

- Components of E and H derived from Maxwell Eqs.



$$e^{-\gamma z} e^{j\omega t}$$

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}:$$

$$\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}:$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & -\gamma \\ E_x & E_y & E_z \end{vmatrix}$$

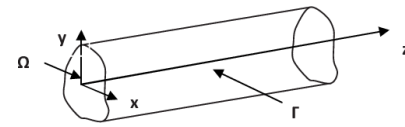
$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & -\gamma \\ H_x & H_y & H_z \end{vmatrix}$$

$$= -j\omega\mu\{\hat{x}H_x + \hat{y}H_y + \hat{z}H_z\}$$

$$= -j\omega\epsilon\{\hat{x}E_x + \hat{y}E_y + \hat{z}E_z\}$$

From $\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}$:	From $\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}$:
$\frac{\partial E_z^0}{\partial y} + \gamma E_y^0 = -j\omega\mu H_x^0 \quad (10-9a)$	$\frac{\partial H_z^0}{\partial y} + \gamma H_y^0 = j\omega\epsilon E_x^0 \quad (10-10a)$
$-\gamma E_x^0 - \frac{\partial E_z^0}{\partial x} = -j\omega\mu H_y^0 \quad (10-9b)$	$-\gamma H_x^0 - \frac{\partial H_z^0}{\partial x} = j\omega\epsilon E_y^0 \quad (10-10b)$
$\frac{\partial E_y^0}{\partial x} - \frac{\partial E_x^0}{\partial y} = -j\omega\mu H_z^0 \quad (10-9c)$	$\frac{\partial H_y^0}{\partial x} - \frac{\partial H_x^0}{\partial y} = j\omega\epsilon E_z^0 \quad (10-10c)$

From $\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}$:	From $\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}$:
$\frac{\partial E_z^0}{\partial y} + \gamma E_y^0 = -j\omega\mu H_x^0 \quad (10-9a)$ $-\gamma E_x^0 - \frac{\partial E_z^0}{\partial x} = -j\omega\mu H_y^0 \quad (10-9b)$	$\frac{\partial H_z^0}{\partial y} + \gamma H_y^0 = j\omega\epsilon E_x^0 \quad (10-10a)$ $-\gamma H_x^0 - \frac{\partial H_z^0}{\partial x} = j\omega\epsilon E_y^0 \quad (10-10b)$



- Components of $\mathbf{E}$ and $\mathbf{H}$ are not all independent.
- Thus, we use E_z and H_z for the expressions of other quantities.

$$E_y = -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial y} - j\omega\mu \frac{\partial H_z}{\partial x} \right)$$

$$H_x = -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial x} - j\omega\epsilon \frac{\partial E_z}{\partial y} \right)$$

$$E_x = -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial x} + j\omega\mu \frac{\partial H_z}{\partial y} \right)$$

$$H_y = -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial y} + j\omega\epsilon \frac{\partial E_z}{\partial x} \right)$$

$$h^2 = k^2 + \gamma^2$$

$$k^2 = \omega^2 \mu \epsilon$$

- Now, the only unknown is E_z and H_z

Discussion on E&H components

- Classify the propagating waves in a uniform waveguide into 3 types:
 - **TEM** waves, neither E_z or H_z exists $H_z=0$, $E_z=0$,
thus **t**ransverse **e**lectro**m**agnetic
 - **TM** waves, E_z exists, $H_z=0$, thus **t**ransverse **m**agnetic
 - **TE** waves, H_z exists, $E_z=0$, thus **t**ransverse **e**lectric

TEM waves

- Since $E_z=H_z=0$, except for the trivial solutions, the denominator 分母 h^2 must be zero

phase velocity

$$\therefore h^2 = k^2 + \gamma^2 = 0 \rightarrow \gamma = jk = j\omega\sqrt{\mu\epsilon} \quad v_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}}$$

- Define the ratio of E_x and H_y as **wave impedance**

$$Z_{TEM} = \frac{E_x}{H_y} = \sqrt{\frac{\mu}{\epsilon}} = \eta$$

From $\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}$:	From $\nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E}$:
$\frac{\partial E_z^0}{\partial y} + \gamma E_y^0 = -j\omega\mu H_x^0 \quad (10-9a)$	$\frac{\partial H_z^0}{\partial y} + \gamma H_y^0 = j\omega\epsilon E_x^0 \quad (10-10a)$
$-\gamma E_x^0 - \frac{\partial E_z^0}{\partial x} = -j\omega\mu H_y^0 \quad (10-9b)$	$-\gamma H_x^0 - \frac{\partial H_z^0}{\partial x} = j\omega\epsilon E_y^0 \quad (10-10b)$
$\frac{\partial E_y^0}{\partial x} - \frac{\partial E_x^0}{\partial y} = -j\omega\mu H_z^0 \quad (10-9c)$	$\frac{\partial H_y^0}{\partial x} - \frac{\partial H_x^0}{\partial y} = j\omega\epsilon E_z^0 \quad (10-10c)$

TEM waves

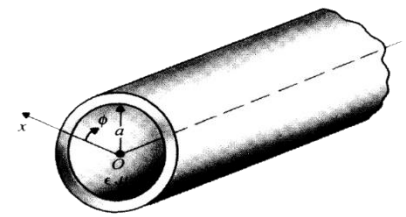
- Similarly, we have

$$Z_{TEM} = -\frac{E_y}{H_x} = \sqrt{\frac{\mu}{\epsilon}} = \eta$$

$$\therefore \vec{H} = -\frac{1}{Z_{TEM}} \vec{a}_z \times \vec{E}$$

- no dispersion for both phase velocity and wave impedance

Discussion



- Single-conductor waveguides cannot support TEM waves
 - Magnetic flux lines always close upon themselves
 - **Ampere circuital law: Line integral** of the magnetic field H around any **closed loop** in a transverse plane must **equal** the **sum of the longitudinal conduction and displacement currents through the loop**.

$$\oint_C \mathbf{H} \cdot d\boldsymbol{\ell} = \int_s \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{s}. \quad (7-54b)$$

Without the inner conductor $\sigma \neq 0$,
no longitudinal conduction current J .

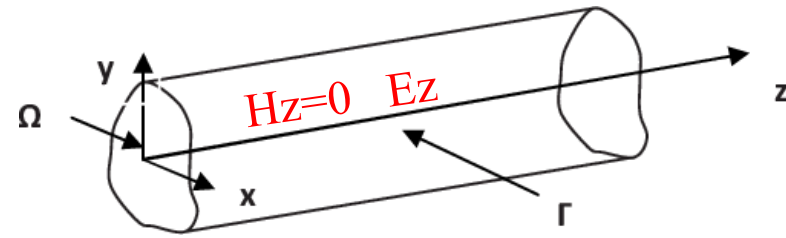
TEM waves does not have z components $E_z = 0$,
no longitudinal displacement current $\partial D / \partial t$

- Thus, **$H=0$** , then **$E=0$**

TM waves

$$\nabla_T^2 \mathbf{E} + h^2 \mathbf{E} = 0$$

$$\nabla_T^2 E_z + h^2 E_z = 0$$



$$h^2 = k^2 + \gamma^2 = k^2 - \beta^2$$

- $H_z=0$

$$H_x = -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial x} - j\omega\epsilon \frac{\partial E_z}{\partial y} \right)$$

$$H_y = -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial y} + j\omega\epsilon \frac{\partial E_z}{\partial x} \right)$$

$$E_x = -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial x} + j\omega\mu \frac{\partial H_z}{\partial y} \right)$$

$$E_y = -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial y} - j\omega\mu \frac{\partial H_z}{\partial x} \right)$$



$$H_x^0 = \frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial y},$$

$$H_y^0 = -\frac{j\omega\epsilon}{h^2} \frac{\partial E_z^0}{\partial x},$$

$$E_x^0 = -\frac{\gamma}{h^2} \frac{\partial E_z^0}{\partial x},$$

$$E_y^0 = -\frac{\gamma}{h^2} \frac{\partial E_z^0}{\partial y}.$$

$$\mathbf{E}_T = -\frac{\gamma}{h^2} \nabla_T E_z$$

$$Z_{TM} = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{\gamma}{j\omega\epsilon}$$

- Note that Z_{TM} is not equals to $j\omega\mu/\gamma$, since for TM waves, γ is not $j\omega\sqrt{\mu\epsilon}$

$$\vec{H} = \frac{1}{Z_{TM}} \vec{a}_z \times \vec{E}$$

TE waves

$$\nabla_T^2 H + h^2 H = 0$$

$$\nabla_T^2 H_z + h^2 H_z = 0$$

$$\begin{aligned} H_x &= -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial x} - j\omega\epsilon \frac{\partial E_z}{\partial y} \right) \\ E_z &= 0 \\ H_y &= -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial y} + j\omega\epsilon \frac{\partial E_z}{\partial x} \right) \\ E_x &= -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial x} + j\omega\mu \frac{\partial H_z}{\partial y} \right) \\ E_y &= -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial y} - j\omega\mu \frac{\partial H_z}{\partial x} \right) \end{aligned}$$

$$\begin{aligned} h^2 &= k^2 + \gamma^2 \\ &= k^2 - \beta^2 \end{aligned}$$

$$H_x = -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial x} - j\omega\epsilon \frac{\partial E_z}{\partial y} \right)$$

$$H_y = -\frac{1}{h^2} \left(\gamma \frac{\partial H_z}{\partial y} + j\omega\epsilon \frac{\partial E_z}{\partial x} \right)$$

$$E_x = -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial x} + j\omega\mu \frac{\partial H_z}{\partial y} \right)$$

$$E_y = -\frac{1}{h^2} \left(\gamma \frac{\partial E_z}{\partial y} - j\omega\mu \frac{\partial H_z}{\partial x} \right)$$



$$H_x^0 = -\frac{\gamma}{h^2} \frac{\partial H_z^0}{\partial x},$$

$$H_y^0 = -\frac{\gamma}{h^2} \frac{\partial H_z^0}{\partial y},$$

$$E_x^0 = -\frac{j\omega\mu}{h^2} \frac{\partial H_z^0}{\partial y},$$

$$E_y^0 = \frac{j\omega\mu}{h^2} \frac{\partial H_z^0}{\partial x}.$$

$$H_T = -\frac{\gamma}{h^2} \nabla_T H_z$$

$$Z_{TE} = \frac{E_x}{H_y} = -\frac{E_y}{H_x} = \frac{j\omega\mu}{\gamma}$$

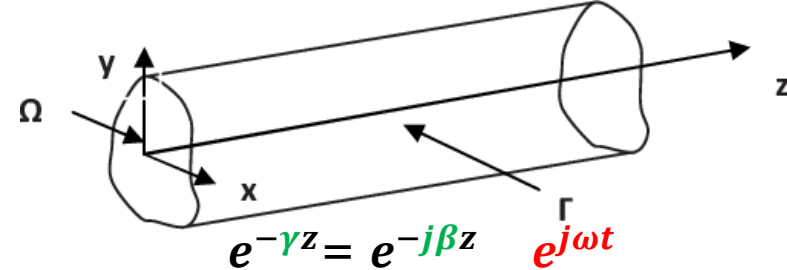
- Note that Z_{TE} is not equals to $\gamma/j\omega\epsilon$, since for TE waves, γ is not $j\omega\sqrt{\mu\epsilon}$

$$\vec{E} = -Z_{TE} (\vec{a}_z \times \vec{H})$$

Finding γ

$$\text{TM} \quad \nabla_T^2 E_z + h^2 E_z = 0$$

$$\text{TE} \quad \nabla_T^2 H_z + h^2 H_z = 0$$



$$h^2 = k^2 + \gamma^2 = k^2 - \beta^2$$

$$k^2 = \omega^2 \mu \epsilon$$

To solve the equations, h can only be some discrete values- eigenvalues

$$\gamma = \sqrt{h^2 - k^2} = jk \sqrt{1 - \frac{h^2}{\omega^2 \mu \epsilon}}$$

$\gamma^2 = 0$ is a natural way to distinguish ranges of the values of the propagation constant.

$$\gamma = 0 \rightarrow h = k = \omega_c \sqrt{\mu \epsilon}$$

$$\omega_c = \frac{h}{\sqrt{\mu \epsilon}} = 2\pi f_c$$

$$\gamma \neq 0 \rightarrow$$

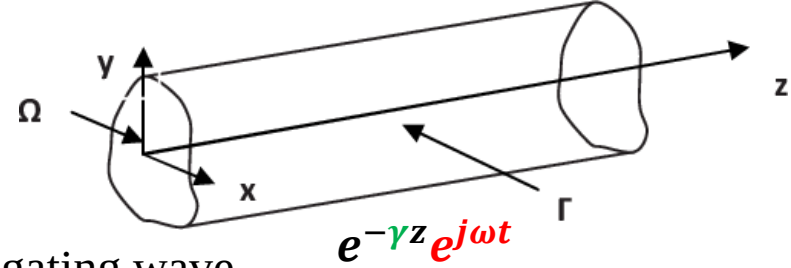
$$\gamma = jk \sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2} = jk \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

$$\gamma^2 < 0, f > f_c$$

or

$$\gamma^2 > 0, f < f_c$$

Propagating wave $\gamma^2 < 0, f > f_c$



$\gamma^2 < 0, \underline{\gamma = j\beta}$ is imaginary, which makes β real, a propagating wave.

$$\gamma = j\beta = jk \sqrt{1 - \left(\frac{\omega_c}{\omega}\right)^2}$$

$$\frac{1}{\lambda_g^2} = \frac{1}{\lambda^2} - \frac{1}{\lambda_c^2}$$

$\lambda_g = \frac{2\pi}{\beta} = \frac{\lambda}{\sqrt{1 - (f_c/f)^2}}$, is the guide wavelength
靠近截止频率, 导波波长无限大

$\lambda_c = \frac{v(=\lambda f)}{f_c} = \frac{1}{\sqrt{\mu\epsilon}f_c}$, is the cutoff wavelength

Phase velocity

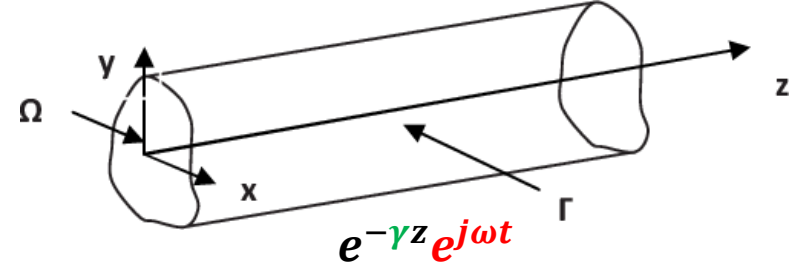
$$\underline{v_p} = \frac{\omega}{\beta} = \frac{\omega \lambda_g}{2\pi} = \frac{v}{\sqrt{1 - (f_c/f)^2}} = \frac{\lambda_g}{\lambda} v > v, v_p \text{ is frequency dependent and always higher than } v$$

group velocity

$$\underline{v_g} = \frac{d\omega}{d\beta} = v \sqrt{1 - (f_c/f)^2} = \frac{\lambda}{\lambda_g} v < v, v_g \text{ is frequency dependent and always lower than } v$$

$$v_p v_g = v^2$$

Evanescent wave $\gamma^2 > 0$, $f < f_c$



$\gamma = \alpha$ is real. Thus all field components contain the propagation factor $e^{-\gamma z} = e^{-\alpha z}$

$$\gamma = \alpha = h \sqrt{1 - \left(\frac{\omega}{\omega_c}\right)^2}$$

- $e^{-\alpha z}$ makes the wave diminishes rapidly with z and is said to be **evanescent**.
- Therefore, a waveguide exhibits the property of a high-pass filter.
- For a given mode, only waves with a frequency higher than the cutoff frequency of the mode can propagate in the guide.

Dispersion relationship

Mode	Guide Wavelength, λ_g
TEM	$\lambda = \frac{1}{f\sqrt{\mu\epsilon}}$
TM	$\frac{\lambda}{\sqrt{1 - (f_c/f)^2}}$
TE	$\frac{\lambda}{\sqrt{1 - (f_c/f)^2}}$

Propagating wave

$d\omega/d\beta$: the group velocity in the waveguide
 ω/β : the phase velocity in the waveguide

